

CK Tile Engine Operation Support Matrix

This matrix shows all CK Tile operations with per-data-type, per-layout, and per-GPU support status. It uses a three-state convention: Y = supported by both CK Tile and tile engine, N = supported by CK Tile but not yet in the tile engine, blank = not supported by CK Tile itself.

Op	CK Tile Kernel	fp16	fp8	bf16	bf8	int8	fp4	fp6	rcr	rrr	ccr	crr	90a	942	950	1201
Activation	softmax	N		N									N	N	N	N
Activation	topk_softmax example: 09_topk_softmax/	N		N									N	N	N	N
Attention	fmha example: 01_fmha/	N	N	N	N								N	N	N	N
Attention	sparse_attn example: 50_sparse_attn/	N		N									N	N	N	N
Conv	grouped_conv example: 20_grouped_convolution/	N		N									N	N	N	N
Data Move	batched_transpose example: 35_batched_transpose/	N	N	N									N	N	N	N
Data Move	image_to_column example: 04_img2col/	N											N	N	N	N
Data Move	permute example: 06_permute/	N	N	N									N	N	N	N
Elementwise	elementwise example: 21_elementwise/	N	N	N	N	N	N						N	N	N	N
GEMM	gemm_multi_d [1] engine: gemm/gemm_multi_d/ example: 19_gemm_multi_d/	Y							Y	Y	Y	Y	Y	Y	Y	
GEMM	gemm_preshuffle [2] engine: gemm/gemm_preshuffle/	Y	Y	Y	Y				Y				Y	Y	Y	N
GEMM	gemm_universal [3] engine: gemm/gemm_universal/ example: 03_gemm/	Y	Y	Y	Y	N			Y	Y	Y	Y	Y	Y	Y	Y
GEMM	streamk_gemm [4] engine: gemm/streamk/ example: 40_streamk_gemm/	Y	Y	Y	Y				Y	Y	Y	Y	Y	Y	N	N
GEMM	batched_contraction example: 41_batched_contraction/	N							N	N	N	N	N	N	N	N

GEMM	batched_gemm example: 16_batched_gemm/	N							N	N	N	N	N	N	N	N
GEMM	block_scale_gemm example: 38_block_scale_gemm/	N	N	N	N		N		N	N	N	N	N	N	N	N
GEMM	flatmm example: 18_flatmm/	N	N	N	N		N	N	N	N	N	N	N	N	N	N
GEMM	gemm_multi_abd example: 22_gemm_multi_abd/	N							N	N	N	N	N	N	N	N
GEMM	gemm_quant						N		N	N	N	N	N	N	N	N
GEMM	grouped_gemm example: 17_grouped_gemm/	N	N	N	N				N	N	N	N	N	N	N	N
GEMM	grouped_gemm_quant															
MoE	fused_moe example: 15_fused_moe/	N	N	N	N	N							N	N	N	N
Norm	add_rmsnorm2d_rdquant example: 11_add_rmsnorm2d_rdquant/	N	N	N	N	N							N	N	N	N
Norm	layernorm2d example: 02_layernorm2d/	N	N	N	N	N							N	N	N	N
Norm	norm_reduce	N		N									N	N	N	N
Norm	rmsnorm2d example: 10_rmsnorm2d/	N	N	N	N	N							N	N	N	N
Pooling	pooling example: 36_pooling/	N											N	N	N	N
Quant	smoothquant example: 12_smoothquant/	N	N	N	N	N							N	N	N	N
Reduce	multi_reduce2d [5] engine: reduce/ example: 05_reduce/	Y		N									N	Y	Y	N
Reduce	reduce2d example: 05_reduce/	N		N									N	N	N	N

Notes:

* [1] * gemm_multi_d: * Builder only accepts * fp16 * . The 4-character layout codes (e.g., * rcr *) include a D tensor layout (always * r *). CMake does not include gfx1201 in * DESIRED_TARGETS * ; gfx1201 has fp16 entries in the shared * GEMM_WARP_TILE_SUPPORTED_COMBINATIONS * but the gemm_multi_d builder does not target it.

* [2] * gemm_preshuffle: * Only layout * rcr * is supported due to preshuffle data format requirements. No dedicated example directory exists. CMake * DESIRED_TARGETS * includes gfx90a, gfx942, and gfx950. The * GEMM_PRESHUFFLE_WARP_TILE_SUPPORTED_COMBINATIONS * dict has entries for fp16, bf16, fp8, and bf8 on gfx90a/gfx942/gfx950 but not gfx1201. Int8 entries exist in the shared preshuffle dict for gfx942 only, but the builder does not accept * int8 * in its * --datatype * choices.

* [3] * gemm_universal: * CMake defaults to * fp8;fp16 * but the builder also accepts * bf16 * and * bf8 * (override with * -DGEMM_UNIVERSAL_DATATYPE *). The * GEMM_WARP_TILE_SUPPORTED_COMBINATIONS * dict has * int8_int8_int32 * entries for gfx942, but the builder does not accept * int8 * — CK Tile's universal GEMM kernel is type-generic so int8

is marked N. gfx1201 only has fp16 tile configurations.

* [4] * streamk_gemm: * CMake defaults to * fp8;fp16 * and layout * rcr * , but the builder accepts all four layouts and also * bf16 * / * bf8 * (override with * -DGEMM_STREAMK_DATATYPE * and * -DGEMM_STREAMK_LAYOUT *). * DESIRED_TARGETS * is * gfx90a;gfx942 * — gfx950 support is pending (TODO in CMakeLists.txt). The builder also lists * fp32 * and * fp64 * in its * --datatype * choices, but no tile configurations or CK Tile example implementations exist for those types. The streamk validation dict has * int8_int8_int32 * for gfx942 but the builder does not accept * int8 * .

* [5] * multi_reduce2d: * The tile engine only supports * fp16 * (per * reduce_parameter.py * TYPE_MAP). CK Tile also implements bf16 (referenced in example code), but the tile engine has no bf16 configurations. * DESIRED_TARGETS * is * gfx942;gfx950 * — gfx90a is not included.

Legend:

- CK Tile Kernel column: First line is the kernel name. Lines prefixed with "engine:" show the tile engine directory under ops/. Lines prefixed with "example:" show the CK Tile example directory under example/ck_tile/.
- Green cell (Y): CK Tile implementation exists and the tile engine supports it.
- Red cell (N): CK Tile implementation exists but the tile engine does not support it.
- Grey cell (blank): No CK Tile implementation exists for this combination.

Layout Codes:

Each layout code is a 3-character string where each character specifies row-major (r) or column-major (c) for tensors A, B, and C respectively. For example, rcr means A=row-major, B=column-major, C=row-major. For gemm_multi_d, a 4-character code is used where the 4th character is the D tensor layout (always r). Layout columns only apply to GEMM-category operations; all other operations leave these columns blank.

Data Type Mapping:

Config Label	A (source)	B (source)	Acc	C (output)
fp16	fp16	fp16	fp32	fp16
bf16	bf16	bf16	fp32	bf16
int8	int8	int8	int32	int32
fp8	fp8	fp8	fp32	fp16
bf8	bf8	bf8	fp32	fp16
fp6	fp6	fp6	fp32	fp32
fp4	fp16/bf16	fp4	fp32	fp16/bf16